

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

2nd Semester of B. Pham Examination

University Theory Examination October -2015

MA 141 Advanced Mathematics

Date: 21.10.2015, Wednesday

Time: 10:00a.m to 01:00p.m

Maximum Marks: 80

Instruction:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION-I

Q-1. Attempt any TWO of the following :

- A (i) Find $L(e^t + 2\cos 3t)$ [02]
(ii) Find Maclaurin's series expansion of $f(x) = \sin x$. [04]
(iii) Using Leibnitz's theorem find $\left(x^2 \cos x\right)_n$. [04]
- B (i) Find 3rd derivative of $\cos 2x$. [02]
(ii) Verify Cauchy's Mean value theorem for the functions $f(x) = x^2, g(x) = x \quad x \in [3, 4]$ [04]
(iii) Find (a) $L^{-1}\left(\frac{4}{s-3}\right)$ (b) $L^{-1}\left(\frac{4s}{s^2+25}\right)$ [04]
- C (i) State Lagrange's mean value theorem. [02]
(ii) Find n^{th} derivative of $x^2 e^x$ using Leibnitz's theorem. [04]
(iii) Using Laplace transformations solve $y'' + 9y = 0, y(0) = 1, y'(0) = 0$ [04]

Q-2. Attempt any TWO of the following :

- A (i) Find (a) $L(\sinh 2t)$ (b) $L(\cos 7t)$ [02]
(ii) Verify Lagrange's mean value theorem for the functions $f(x) = x^2 + 1$ in $[0, 2]$. [04]
(iii) Find Taylor's series expansion for $x^3 - 2x + 4$ in power of $(x - 2)$. [04]
- B (i) State Rolle's theorem. [02]
(ii) Using Laplace transforms solve the equation $y'' + y = 0, y(0) = A$ and $y'(0) = B$. [04]
(iii) $L\left(4e^t - 4t^2 + \cos 7t + \frac{7}{2}\sinh 2t\right)$ [04]
- C (i) If $y = e^{2x} \sin(3x + 5)$ then $y_n =$ [02]
(ii) Using Leibnitz's theorem find $(x \sin 3x)_n$. [04]
(iii) Write Maclaurin's series expansion of $f(x) = e^x$. [04]

SECTION-II

Q-3. Attempt any TWO of the following :

- A**
- (i) Find general solution of $y'' = 0$ [02]
 - (ii) Using Exact equation solve $2xydx + (x^2 + y^2)dy = 0$. [04]
 - (iii) Bacteria at the rate proportional to the number present. If the original number N doubles in three hours, find in how many hours, the numbers of the bacteria will be $4N$. [04]
- B**
- (i) Find general solution of $y'' + 4y' - 5y = 0$ [02]
 - (ii) Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \sec x$ [04]
 - (iii) Show that $y = ae^x + be^{2x}$ is a solution of $y'' - 3y' + 2y = 0$. [04]
- C**
- (i) write the characteristic equation for $2y'' + 2y' + 3y = 0$ [02]
 - (ii) Show that $y = ax^3 + bx^2$ is a solution of $x^2y'' - 4xy' + 6y = 0$. [04]
 - (iii) Using Exact equation solve $\sin y dx + x \cos y dy$. [04]

Q-4. Attempt any TWO of the following :

- A**
- (i) Find order and degree of $\left(\frac{d^2y}{dx^2}\right)^3 + 14y = x$ [02]
 - (ii) Using Linear equation solve $\frac{dy}{dx} - 4xy = x^3$. [04]
 - (iii) Find general solution of (a) $y'' + 8y' - 16y = 0$ (b) $y'' - 2y' + 6y = 0$ [04]
- B**
- (i) Find order and degree of $\left(\frac{d^3y}{dx^3}\right)^2 = \sin 3x + e^x$ [02]
 - (ii) Find general solution of (a) $y'' + 2y' + y = 0$ (b) $y'' - 6y' + 25y = 0$ [04]
 - (iii) Using Linear equation solve $\frac{dy}{dx} - 2y = x$. [04]
- C**
- (i) Write the general form of Exact equations of first order and first degree. [02]
 - (ii) The rate at which bacteria multiply is proportional to the instantaneous number present. If the original number doubles in 2 hours, in how many hours will it triple? [04]
 - (iii) Using Linear equation solve $\frac{dy}{dx} + 2y = e^{3x}$. [04]
